

H3 Definite integrals

Name: _____

Class: _____

Date: _____

Time: **420 minutes**

Marks: **348 marks**

Comments:

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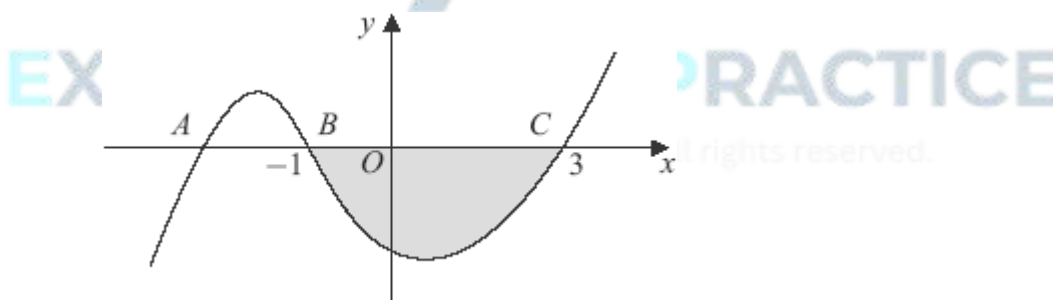
Q1.

- (a) Use integration by parts to find $\int (t - 1) \ln t \, dt$. (4)
- (b) Use the substitution $t = 2x + 1$ to show that $\int 4x \ln(2x + 1) \, dx$ can be written as $\int (t - 1) \ln t \, dt$. (3)
- (c) Hence find the exact value of $\int_0^1 4x \ln(2x + 1) \, dx$. (3)

(Total 10 marks)

Q2.

- (a) The polynomial $p(x)$ is given by $p(x) = x^3 - 7x - 6$.
- (i) Use the Factor Theorem to show that $x + 1$ is a factor of $p(x)$. (2)
- (ii) Express $p(x) = x^3 - 7x - 6$ as the product of three linear factors. (3)
- (b) The curve with equation $y = x^3 - 7x - 6$ is sketched below.



The curve cuts the x -axis at the point A and the points $B(-1, 0)$ and $C(3, 0)$.

- (i) State the coordinates of the point A . (1)
- (ii) Find $\int_{-1}^3 (x^3 - 7x - 6) \, dx$. (5)
- (iii) Hence find the area of the shaded region bounded by the curve $y = x^3 - 7x - 6$ and the x -axis between B and C .

(1)

(iv) Find the gradient of the curve $y = x^3 - 7x - 6$ at the point B .

(3)

(v) Hence find an equation of the normal to the curve at the point B .

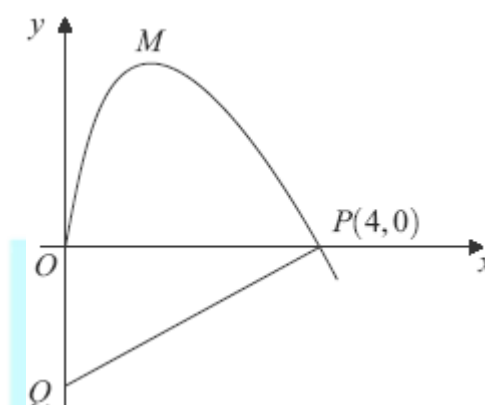
(3)

(Total 18 marks)

Q3.

A curve, drawn from the origin O , crosses the x -axis at the point $P(4, 0)$.

The normal to the curve at P meets the y -axis at the point Q , as shown in the diagram.



The curve, defined for $x \geq 0$, has equation

$$y = 4x^{\frac{1}{2}} - x^{\frac{3}{2}}$$

(a) (i) Find $\frac{dy}{dx}$.

(3)

(ii) Show that the gradient of the curve at $P(4, 0)$ is -2 .

(2)

(iii) Find an equation of the normal to the curve at $P(4, 0)$.

(3)

(iv) Find the y -coordinate of Q and hence find the area of triangle OPQ .

(3)

(v) The curve has a maximum point M . Find the x -coordinate of M .

(3)

(b) (i) Find $\int \left(4x^{\frac{1}{2}} - x^{\frac{3}{2}} \right) dx$.

(3)

- (ii) Find the total area of the region bounded by the curve and the lines PQ and QO .

(3)

(Total 20 marks)

Q4.

(a) (i) Given that $y = 2x^2 - 8x + 3$, find $\frac{dy}{dx}$.

(1)

- (ii) Hence, or otherwise, find

$$\int_4^6 \frac{x-2}{2x^2-8x+3} dx$$

giving your answer in the form $k \ln 3$, where k is a rational number.

(4)

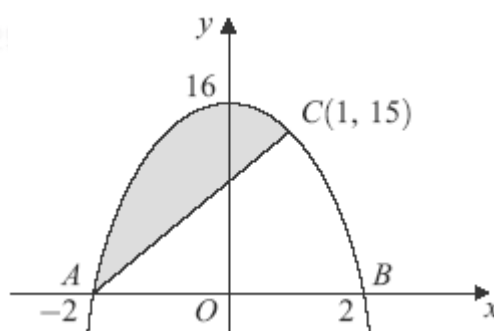
- (b) Use the substitution $u = 3x - 1$ to find , giving your answer in terms of x .

(4)

(Total 9 marks)

Q5.

The curve with equation $y = 16 - x^4$ is sketched below.



The points $A(-2, 0)$, $B(2, 0)$ and $C(1, 15)$ lie on the curve.

- (a) Find an equation of the straight line AC .

(3)

(b) (i) Find $\int_{-2}^1 (16 - x^4) dx$.

(5)

- (ii) Hence calculate the area of the shaded region bounded by the curve and the line AC .

(3)

(Total 11 marks)

Q6.

- (a) The expression $\left(1 + \frac{4}{x^2}\right)^3$ can be written in the form

$$1 + \frac{p}{x^2} + \frac{q}{x^4} + \frac{64}{x^6}$$

By using the binomial expansion, or otherwise, find the values of the integers p and q .

(3)

- (b) (i) Hence find $\int \left(1 + \frac{4}{x^2}\right)^3 dx$.

(4)

- (ii) Hence find the value of $\int_1^2 \left(1 + \frac{4}{x^2}\right)^3 dx$.

(2)

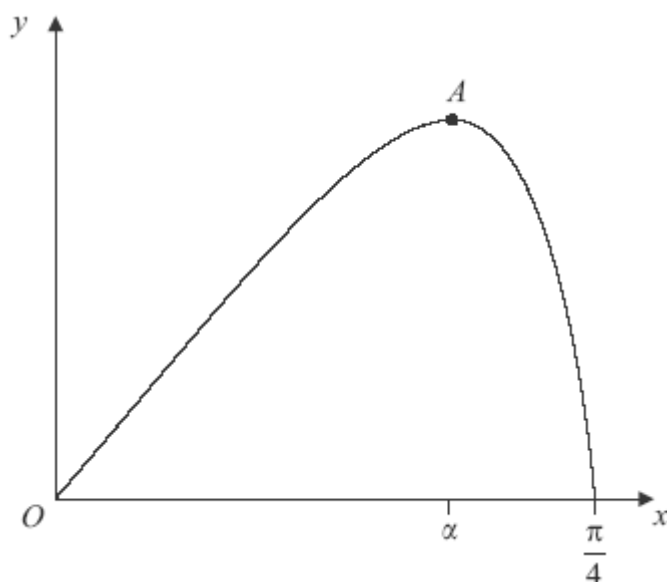
(Total 9 marks)

Q7.

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A curve is defined for $0 \leq x \leq \frac{\pi}{4}$ by the equation $y = x \cos 2x$, and is sketched below.

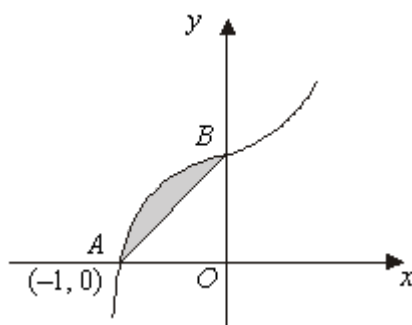


- (a) Find $\frac{dy}{dx}$. (2)
- (b) The point A , where $x = \alpha$, on the curve is a stationary point.
- (i) Show that $1 - 2\alpha \tan 2\alpha = 0$. (2)
- (ii) Show that $0.4 < \alpha < 0.5$. (2)
- (iii) Show that the equation $1 - 2x \tan 2x = 0$ can be rearranged to become $x = \frac{1}{2} \tan^{-1}\left(\frac{1}{2x}\right)$. (1)
- (iv) Use the iteration $x_{n+1} = \frac{1}{2} \tan^{-1}\left(\frac{1}{2x_n}\right)$ with $x_1 = 0.4$ to find x_3 , giving your answer to two significant figures. (2)
- (c) Use integration by parts to find $\int_0^{0.5} x \cos 2x dx$, giving your answer to three significant figures. (5)

(Total 14 marks)

Q8.

The curve with equation $y = 3x^5 + 2x + 5$ is sketched below.



The curve cuts the x -axis at the point $A(-1, 0)$ and cuts the y -axis at the point B .

- (a) (i) State the coordinates of the point B and hence find the area of the triangle AOB , where O is the origin.

(3)

- (ii) Find $\int (3x^5 + 2x + 5) dx$

(3)

- (iii) Hence find the area of the shaded region bounded by the curve and the line AB .

(4)

- (b) (i) Find the gradient of the curve with equation $y = 3x^5 + 2x + 5$ at the point $A(-1, 0)$.

(3)

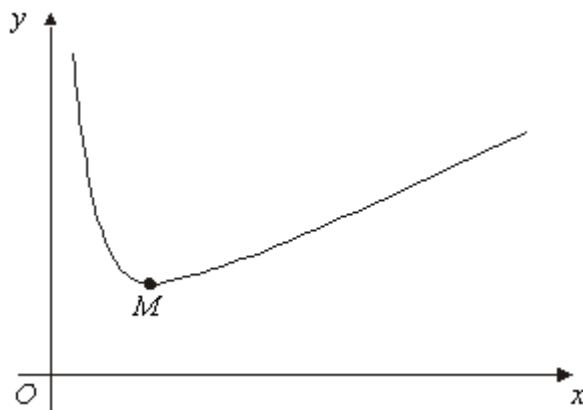
- (ii) Hence find an equation of the tangent to the curve at the point A .

(1)

(Total 14 marks)

Q9.

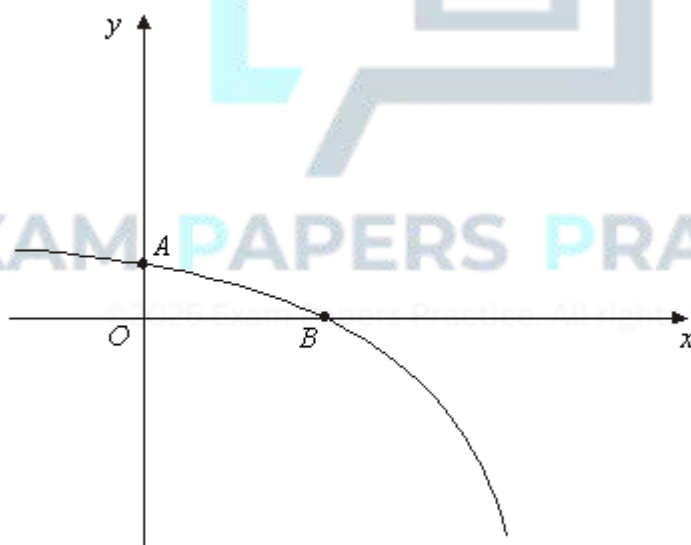
A curve C is defined for $x > 0$ by the equation $y = x + 1 + \frac{4}{x^2}$ and is sketched below.



- (a) (i) Given that $y = x + 1 + \frac{4}{x^2}$, find $\frac{dy}{dx}$. (3)
- (ii) The curve C has a minimum point M . Find the coordinates of M . (4)
- (iii) Find an equation of the normal to C at the point $(1, 6)$. (4)
- (b) (i) Find $\int \left(x + 1 + \frac{4}{x^2} \right) dx$. (3)
- (ii) Hence find the area of the region bounded by the curve C , the lines $x = 1$ and $x = 4$ and the x -axis. (2)
- (Total 16 marks)

Q10.

The sketch shows the graph of $y = 4 - e^{2x}$. The curve crosses the y -axis at the point A and the x -axis at the point B .

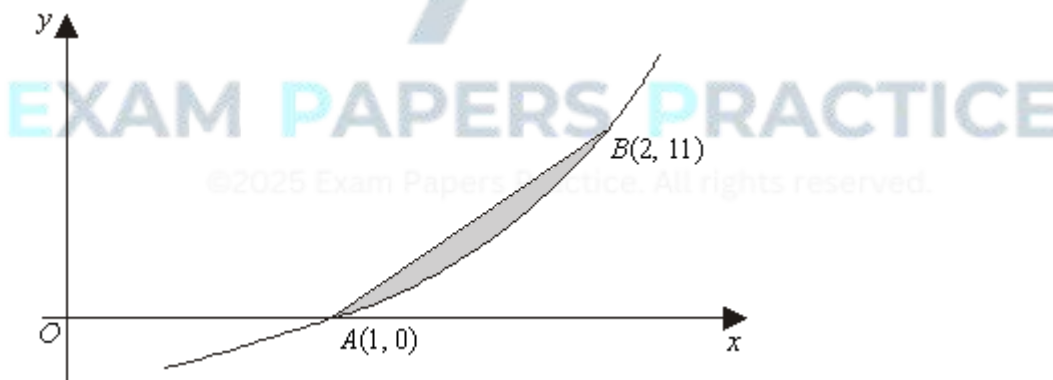


- (a) (i) Find $\int (4 - e^{2x}) dx$. (2)
- (ii) Hence show that $\int_0^{\ln 2} (4 - e^{2x}) dx = 4 \ln 2 - \frac{3}{2}$. (2)
- (b) (i) Write down the y -coordinate of A .

- (1)
- (ii) Show that $x = \ln 2$ at B . (2)
- (c) Find the equation of the normal to the curve $y = 4 - e^{2x}$ at the point B . (4)
- (d) Find the area of the region enclosed by the curve $y = 4 - e^{2x}$, the normal to the curve at B and the y -axis. (3)
- (Total 14 marks)

Q11.

- (a) The polynomial $f(x)$ is given by $f(x) = x^3 + 4x - 5$.
- (i) Use the Factor Theorem to show that $x - 1$ is a factor of $f(x)$. (2)
- (ii) Express $f(x)$ in the form $(x - 1)(x^2 + px + q)$, where p and q are integers. (2)
- (iii) Hence show that the equation $f(x) = 0$ has exactly one real root and state its value. (3)
- (b) The curve with equation $y = x^3 + 4x - 5$ is sketched below.



The curve cuts the x -axis at the point $A(1, 0)$ and the point $B(2, 11)$ lies on the curve.

- (i) Find $\int (x^3 + 4x - 5) dx$. (3)
- (ii) Hence find the area of the shaded region bounded by the curve and the line AB . (4)

(Total 14 marks)

Q12.

(a) Simplify:

(i) $x^{\frac{3}{2}} \times x^{\frac{1}{2}};$ (1)

(ii) $x^{\frac{3}{2}} \div x;$ (1)

(iii) $\left(x^{\frac{3}{2}}\right)^2$ (1)

(b) (i) Find $\int 3x^{\frac{1}{2}} dx.$ (3)

(ii) Hence find the value of $\int_1^9 3x^{\frac{1}{2}} dx.$ (2)
 (Total 8 marks)

Q13.

(a) Use integration by parts to find $\int xe^{5x} dx.$ (4)

(b) (i) Use the substitution $u = \sqrt{x}$ to show that

$$\int \frac{1}{\sqrt{x}(1+\sqrt{x})} dx = \int \frac{2}{1+u} du$$

(2)

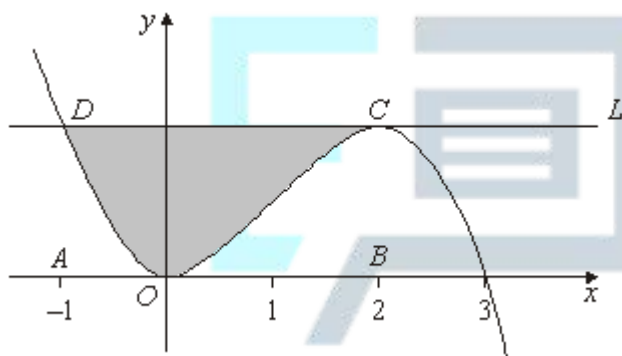
(ii) Find the exact value of $\int_1^9 \frac{1}{\sqrt{x}(1+\sqrt{x})} dx$ (3)
 (Total 9 marks)

Q14.

- (a) Write down $\int \sec^2 x \, dx$. (1)
- (b) Given that $y = \frac{\cos x}{\sin x}$, use the quotient rule to show that $\frac{dy}{dx} = -\operatorname{cosec}^2 x$. (4)
- (c) Prove the identity $(\tan x + \cot x)^2 = \sec^2 x + \operatorname{cosec}^2 x$. (3)
- (d) Hence find $\int_{0.5}^1 (\tan x + \cot x)^2 \, dx$ giving your answer to two significant figures. (4)
- (Total 12 marks)

Q15.

The diagram shows the curve with equation $y = 3x^2 - x^3$ and the line L .



The points A and B have coordinates $(-1, 0)$ and $(2, 0)$ respectively. The curve touches the x -axis at the origin O and crosses the x -axis at the point $(3, 0)$. The line L cuts the curve at the point D where $x = -1$ and touches the curve at C where $x = 2$.

- (a) Find the area of the rectangle $ABCD$. (2)
- (b) (i) Find $\int (3x^2 - x^3) \, dx$. (3)
- (ii) Hence find the area of the shaded region bounded by the curve and the line L . (4)
- (c) For the curve above with equation $y = 3x^2 - x^3$:
- (i) find $\frac{dy}{dx}$;

(2)

(ii) hence find an equation of the tangent at the point on the curve where $x = 1$;

(3)

(iii) show that y is decreasing when $x^2 - 2x > 0$.

(2)

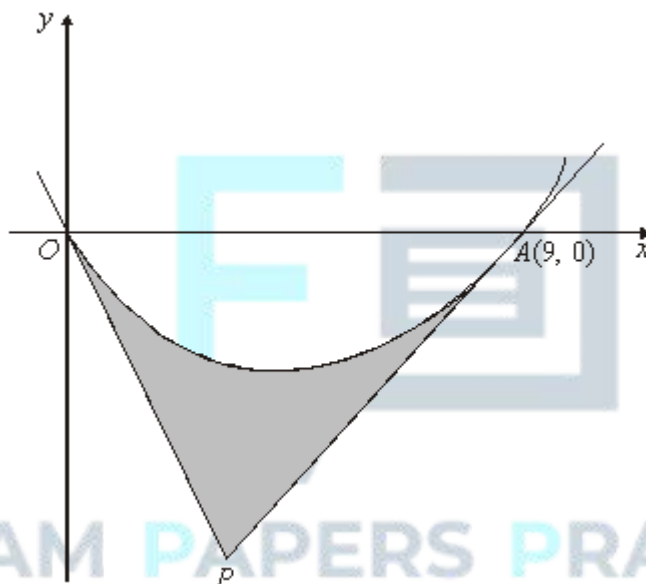
(d) Solve the inequality $x^2 - 2x > 0$.

(2)

(Total 18 marks)

Q16.

A curve, drawn from the origin O , crosses the x -axis at the point $A(9, 0)$. Tangents to the curve at O and A meet at the point P , as shown in the diagram.



The curve, defined for $x \geq 0$, has equation

$$y = x^{\frac{3}{2}} - 3x$$

(a) Find $\frac{dy}{dx}$.

(2)

(b) (i) Find the value of $\frac{dy}{dx}$ at the point O and hence write down an equation of the tangent at O .

(2)

(ii) Show that the equation of the tangent at $A(9, 0)$ is $2y = 3x - 27$.

(3)

(iii) Hence find the coordinates of the point P where the two tangents meet.

(3)

(c) Find $\int \left(x^{\frac{3}{2}} - 3x \right) dx$.

(3)

(d) Calculate the area of the shaded region bounded by the curve and the tangents OP and AP .

(5)

(Total 18 marks)

Q17.

(a) (i) Given that $f(x) = x^4 + 2x$, find $f'(x)$.

(1)

(ii) Hence, or otherwise, find $\int \frac{2x^3 + 1}{x^4 + 2x} dx$.

(2)

(b) (i) Use the substitution $u = 2x + 1$ to show that

$$\int x\sqrt{2x+1} dx = \frac{1}{4} \int \left(u^{\frac{3}{2}} - u^{\frac{1}{2}} \right) du$$

(3)

(ii) Hence show that $\int_0^4 x\sqrt{2x+1} dx = 19.9$ correct to three significant figures.

(4)

(Total 10 marks)

Q18.

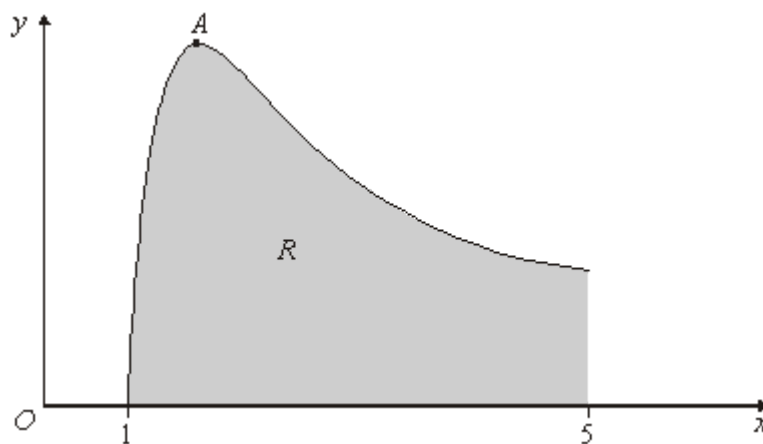
(a) Given that $y = x^{-2} \ln x$, show that $\frac{dy}{dx} = \frac{1 - 2 \ln x}{x^3}$.

(4)

(b) Using integration by parts, find $\int x^{-2} \ln x dx$.

(4)

(c) The sketch shows the graph of $y = x^{-2} \ln x$.



- (i) Using the answer to part (a), find, in terms of e , the x -coordinate of the stationary point A . (2)
- (ii) The region R is bounded by the curve, the x -axis and the line $x = 5$. Using your answer to part (b), show that the area of R is

$$\frac{1}{5} (4 - \ln 5)$$

(3)
(Total 13 marks)

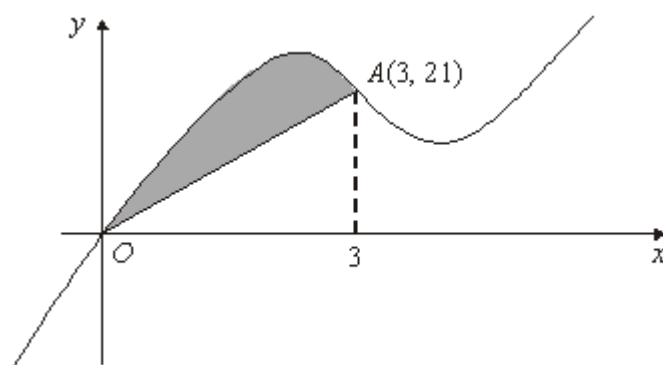
Q19.

- (a) Express $\cos 2x$ in the form $a \cos^2 x + b$, where a and b are constants. (2)

- (b) Hence show that $\int_0^{\frac{\pi}{2}} \cos^2 x \, dx = \frac{\pi}{a}$, where a is an integer. (5)
- (Total 7 marks)

Q20.

The curve with equation $y = x^3 - 10x^2 + 28x$ is sketched below.



The curve crosses the x -axis at the origin O and the point $A(3, 21)$ lies on the curve.

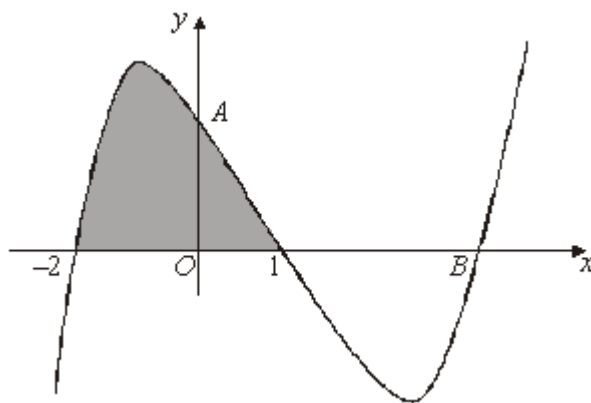
- (a) (i) Find $\frac{dy}{dx}$. (3)
- (ii) Hence show that the curve has a stationary point when $x = 2$ and find the x -coordinate of the other stationary point. (4)
- (b) (i) Find $\int (x^3 - 10x^2 + 28x) dx$. (3)
- (ii) Hence show that $\int_0^3 (x^3 - 10x^2 + 28x) dx = 56\frac{1}{4}$. (2)
- (iii) Hence determine the area of the shaded region bounded by the curve and the line OA . (3)
- (Total 15 marks)**

Q21.

- (a) Given that $y = x \ln x$, find $\frac{dy}{dx}$. (2)
- (b) Hence, or otherwise, find $\int_1^{\ln} x \, dx$. (2)
- (c) Find the exact value of $\int_1^5 \frac{1}{x} \, dx$. (2)
- (Total 6 marks)**

Q22.

- (a) The function f is defined for all values of x by $f(x) = x^3 - 3x^2 - 6x + 8$.
- (i) Given that $f(1) = 0$ and $f(-2) = 0$, write down two linear factors of $f(x)$. (2)
- (ii) Hence express $x^3 - 3x^2 - 6x + 8$ as the product of three linear factors. (2)
- (b) The curve with equation $y = x^3 - 3x^2 - 6x + 8$ is sketched below.



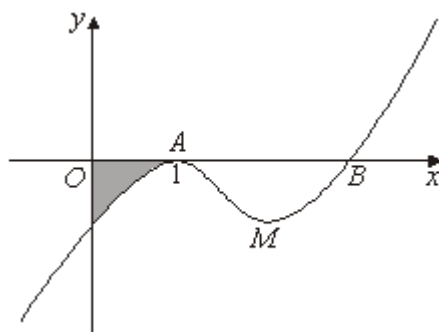
- (i) The curve intersects the y -axis at the point A . Find the y -coordinate of A . (1)
- (ii) The curve crosses the x -axis when $x = -2$, when $x = 1$ and also at the point B . Use the results from part (a) to find the x -coordinate of B . (1)
- (c) (i) Find $\int (x^3 - 3x^2 - 6x + 8) dx$ (4)
- (ii) Hence find the area of the shaded region bounded by the curve and the x -axis. (3)
- (Total 13 marks)**

Q23.

- (a) Write $\sqrt{x}$ in the form x^k , where k is a fraction. (1)
- (b) Hence express $\sqrt{x}(x-1)$ in the form $x^p - x^q$. (2)
- (c) Find $\int \sqrt{x}(x-1) dx$. (3)
- (d) Hence show that $\int_1^2 \sqrt{x}(x-1) dx = \frac{4}{15}(\sqrt{2} + 1)$. (3)
- (Total 9 marks)**

Q24.

The curve with equation $y = x^3 - 5x^2 + 7x - 3$ is sketched below.



The curve touches the x -axis at the point $A(1, 0)$ and cuts the x -axis at the point B .

- (a) (i) Use the factor theorem to show that $x - 3$ is a factor of

$$p(x) = x^3 - 5x^2 + 7x - 3$$

(2)

- (ii) Hence find the coordinates of B .

(1)

- (b) The point M , shown on the diagram, is a minimum point of the curve with equation $y = x^3 - 5x^2 + 7x - 3$.

- (i) Find $\frac{dy}{dx}$.

(2)

- (ii) Hence determine the x -coordinate of M .

(3)

- (c) Find the value of $\frac{d^2y}{dx^2}$ when $x = 1$.

(2)

- (d) (i) Find $\int (x^3 - 5x^2 + 7x - 3) dx$.

(4)

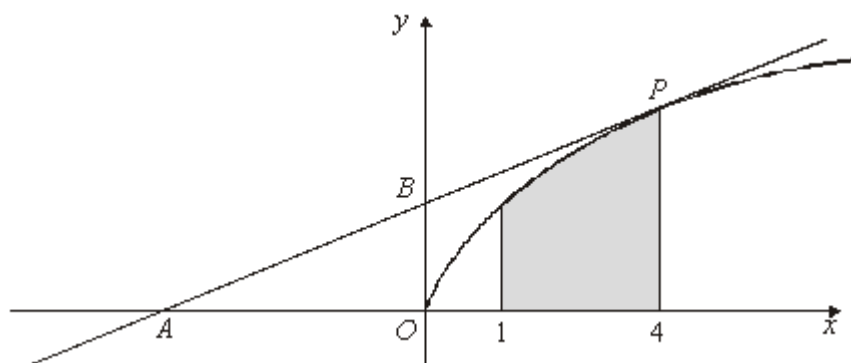
- (ii) Hence determine the area of the shaded region bounded by the curve and the coordinate axes.

(4)

(Total 18 marks)

Q25.

The diagram shows a curve C with equation $y = \sqrt{x}$. The point O is the origin $(0, 0)$.

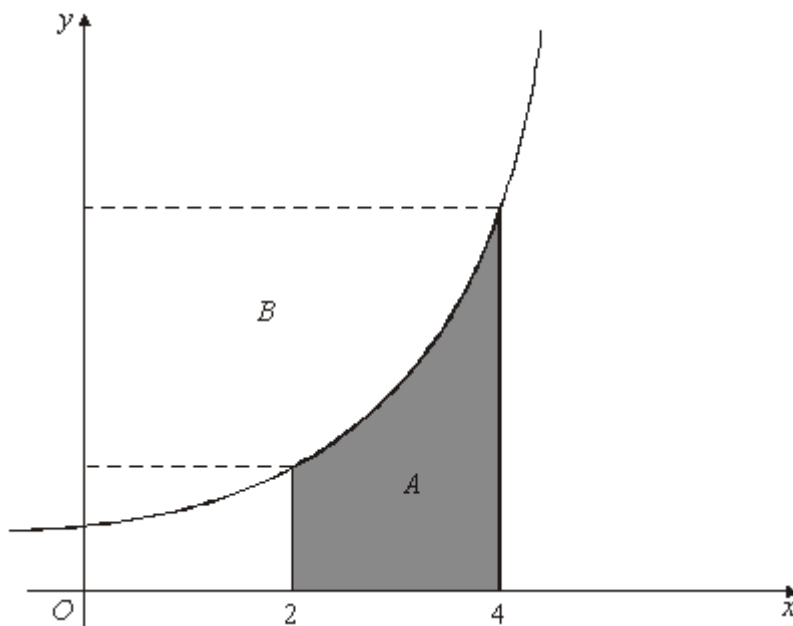


The region bounded by the curve C , the x -axis and the vertical lines $x = 1$ and $x = 4$ is shown shaded in the diagram.

- (a) (i) Write $\sqrt{x}$ in the form x^p , where p is a constant. (1)
- (ii) Find $\int \sqrt{x} \, dx$. (2)
- (iii) Hence find the area of the shaded region. (3)
- (b) The point on C for which $x = 4$ is P . The tangent to C at the point P intersects the x -axis and the y -axis at the points A and B respectively.
- (i) Find an equation for the tangent to the curve C at the point P . (4)
- (ii) Find the area of the triangle AOB . (3)
- (c) Describe the single geometrical transformation by which the curve with equation $y = \sqrt{x-1}$ can be obtained from the curve C . (2)
- (d) Use the trapezium rule with four ordinates (three strips) to find an approximation for $\int_1^4 \sqrt{x-1} \, dx$, giving your answer to three significant figures. (4)
- (Total 19 marks)

Q26.

The diagram shows part of the graph of $y = e^{x^2} + 3$.



- (a) Describe a sequence of two geometrical transformations that maps the graph of $y = e^x$ onto the graph of $y = e^{2x} + 3$. (4)
- (b) Find the exact value of the area of the shaded region A . (4)
- (c) The region B is indicated on the diagram.
- Find the area of the region B , giving your answer in the form $pe^8 + qe^4$, where p and q are numbers to be determined. (4)

(Total 12 marks)

Q27.

- (a) Express $\sin 2x$ in terms of $\sin x$ and $\cos x$. (1)
- (b) Using the identity $\cos(A + B) = \cos A \cos B - \sin A \sin B$:
- (i) express $\cos 2x$ in terms of $\sin x$ and $\cos x$; (2)
- (ii) show, by writing $3x$ as $(2x + x)$, that
- $$\cos 3x = 4 \cos^3 x - 3 \cos x$$
- (4)

- (c) Show that $\int_0^{\frac{\pi}{2}} \cos^3 x \, dx = \frac{2}{3}$.

(5)
(Total 12 marks)



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